$All\ formulae$

1) $Simple\ linear\ regression: \ y = b_0 + b_1 * x_1$
2) $Multiple\ linear\ regression: y = b_0 + b_1 * x_1 + b_2 * x_2 + \cdots + b_n * x_n$
3) $error: (y_a - y_p) = Residual$
4) $SSE: \sum_{i=1}^{n} (\boldsymbol{y_{a_i}} - \boldsymbol{y_{p_i}})^2 = \boldsymbol{RSE}$
5) $MSE: \frac{1}{N} * \sum_{i=1}^{n} (\boldsymbol{y_{a_i}} - \boldsymbol{y_{p_i}})^2$
6) $RMSE: \sqrt{\frac{\boldsymbol{1}}{\boldsymbol{N}} * \sum_{i=1}^{n} (\boldsymbol{y_{a_i}} - \boldsymbol{y_{p_i}})^2}$
7) $Loss : (y_a - y_p)$
8) $Cost : j = \frac{1}{N} * \sum_{i=1}^{n} (\boldsymbol{y_{a_i}} - \boldsymbol{y_{p_i}})^2$
9) $b_o = \overline{y_a} - b_1 * \bar{x}$

$$b_1 = \frac{\sum_i^n (\overline{y_a} - \boldsymbol{y_{ai}}) * (\bar{x} - \boldsymbol{x_i})}{\sum_i^n (\bar{x} - \boldsymbol{x_i})^2} = \frac{Cov(x,y)}{Var(x)}$$

10)
11) $Correlation(r) = \frac{Cov(x,y)}{\sigma_x * \sigma_y}$
12) $RSE = \sqrt{\frac{RSS}{n-p-1}} = \sqrt{\frac{1}{n-p-1} \sum_{i=1}^{n} (y_{actual} - y_{predicted})^2}$
13) $R - squared = 1 - \frac{sum\ of\ squared\ residuals\ (SSR)}{Total\ sum\ of\ squares (SST)}$
14) $R - squared = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$
15) $R_{Adj} = 1 - \frac{[1 - R_{square}] * (N-1)}{N-p-1}$
16) $AIC : \quad 2P - 2\ln(L)$
17) $BIC : \quad p * \ln(N) - 2\ln(L)$
18) $VIF = \frac{1}{1 - R_{Square}}$
19) $DW = \frac{\sum_{t=2}^{n} (e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2}$
20) $Rige\ cost\ function = \frac{1}{n} * \sum_{i}^{n} (y_a - y_p)^2 + \lambda * \sum_i^n b_i^2$
21) $Lasso\ cost\ function = \frac{1}{n} * \sum_{i}^{n} (y_a - y_p)^2 + \lambda * \sum_i^n |b_i|$